

Two-Step Equations in ContextWorksheet 7A.01.04 **ANSWER KEY**

Frankie solved the equation $6y + 2.5 = 46$. Her work is shown. Use it for questions 1 and 2.

$$\begin{aligned} 6y + 2.5 - 2.5 &= 46 - 2.5 \\ 6y &= 43.5 \\ 6y \div 6 &= 43.5 \div 6 \\ y &= 7.25 \end{aligned}$$

1. Use substitution to decide whether Frankie's solution is correct.

$$6(7.25) + 2.5 = 43.5 + 2.5 = 46 \checkmark$$

Her solution is correct.

2. If Frankie is correct, name the properties of equality she used. If she is not, describe her error.

Correct. She used the subtraction property of equality, then the division property of equality.

Multiplying both sides by $1/6$ instead would be the multiplication property, and is equally correct.

Alishia solved the equation $7x - 4\frac{1}{4} = 44.75$. Her work is shown. Use it for questions 3 and 4.

$$\begin{aligned} 7x - 4\frac{1}{4} - 4\frac{1}{4} &= 44.75 - 4\frac{1}{2} \\ 7x &= 40.25 \\ 7x \div 7 &= 40.25 \div 7 \\ x &= 5.75 \end{aligned}$$

3. Use substitution to decide whether Alishia's solution is correct.

$$7(5.75) - 4.25 = 40.25 - 4.25 = 36$$

$36 \neq 44.75$, so her solution is not correct.

4. Describe the errors Alishia made, then solve the equation correctly.

She made two errors. First, she subtracted $4\frac{1}{4}$ from the left side when she needed to add it — the left side already has $4\frac{1}{4}$ subtracted. Second, she used $4\frac{1}{4}$ on the left but $4\frac{1}{2}$ on the right, so the two sides no longer match.

$$7x - 4.25 + 4.25 = 44.75 + 4.25$$

$$7x = 49$$

$$x = 7 \quad \text{Check: } 7(7) - 4.25 = 49 - 4.25 = 44.75 \checkmark$$

A student who names only one of the two errors has not answered the question, which asks for the errors. Worth a follow-up question at the desk rather than the point.

For questions 5 and 6, solve the equation, then say what the solution means in the situation.

5. A gym charges a \$55 joining fee plus a monthly fee. A full year costs \$235. The equation $55 + 12t = 235$ represents the situation.

$$12t = 180 \quad t = 15$$

The monthly fee is \$15.

Check: $55 + 12(15) = 55 + 180 = 235$. The variable t is a monthly rate, not a number of months.

6. Renting a kayak costs \$24 for the first 3 hours and \$4.50 for each additional hour. You spent \$42. The equation $24 + 4.50m = 42$ represents the situation.

$$4.50m = 18 \quad m = 4$$

You kayaked for 4 additional hours, so 7 hours in all.

The most common wrong answer is "4 hours." The variable counts hours BEYOND the first three.

For questions 7 through 10, solve the equation and check your solution by substitution.

7. $8 + \frac{z}{4} = 23$

$$8 + z/4 - 8 = 23 - 8$$

$$z/4 = 15$$

$$(z/4) \times 4 = 15 \times 4$$

$$z = 60$$

$$\text{Check: } 8 + 60 \div 4 = 8 + 15 = 23 \checkmark$$

Subtraction property, then multiplication property.

8. $5t - 17.2 = 16.3$

$$5t - 17.2 + 17.2 = 16.3 + 17.2$$

$$5t = 33.5$$

$$5t \div 5 = 33.5 \div 5$$

$$t = 6.7$$

$$\text{Check: } 5(6.7) - 17.2 = 33.5 - 17.2 = 16.3 \checkmark$$

Addition property, then division property.

9. $\frac{s}{3} - 0.6 = 1.2$

$$s/3 - 0.6 + 0.6 = 1.2 + 0.6$$

$$s/3 = 1.8$$

$$(s/3) \times 3 = 1.8 \times 3$$

$$s = 5.4$$

$$\textbf{Check: } 5.4 \div 3 - 0.6 = 1.8 - 0.6 = 1.2 \checkmark$$

Addition property, then multiplication property.

10. $2.4a + 8 = 27.2$

$$2.4a + 8 - 8 = 27.2 - 8$$

$$2.4a = 19.2$$

$$2.4a \div 2.4 = 19.2 \div 2.4$$

$$a = 8$$

$$\textbf{Check: } 2.4(8) + 8 = 19.2 + 8 = 27.2 \checkmark$$

Subtraction property, then division property.